Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Trigonometry

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Solve $\sin x = \frac{1}{2}$ for $0 \le x \le 2\pi$. **[3]**

2. Solve $\cos x = \frac{\sqrt{3}}{2}$ for $0 \le x \le 2\pi$. **[3]**

3. Solve $\tan x = 1$ for $0 \le x \le 2\pi$. **[3]**

4. Solve $\sin x = -\frac{\sqrt{2}}{2}$ for $0 \le x \le 2\pi$. **[3]**

5. Solve $2\cos x - 1 = 0$ for $0 \le x \le 2\pi$. **[3]**

6. Solve $2\sin x + \sqrt{3} = 0$ for $0 \le x \le 2\pi$. **[3]**

7. Prove $\sin^2\theta + \cos^2\theta = 1$. **[2]**

8. Prove $1 + \tan^2\theta = \sec^2\theta$. **[3]**

9. Prove $\sin 2\theta = 2\sin\theta\cos\theta$. **[3]**

10. Prove $\cos 2\theta = \cos^2\theta - \sin^2\theta$. **[3]**

11. Solve $\sin 2x = \frac{1}{2}$ for $0 \le x \le \pi$. **[4]**

12. Solve $\cos 2x = 0$ for $0 \leq x \leq \pi$. **[3]**

13. Find the exact value of $\sin 15°$. **[4]**

14. Find the exact value of $\cos 75°$. **[4]**

15. Solve $2\sin^2 x - \sin x = 0$ for $0 \leq x \leq 2\pi$. **[4]**

Section B – Hard (Q16–25)

16. Solve $2\cos^2 x + \cos x - 1 = 0$ for $0 \leq x \leq 2\pi$. **[5]**

17. Solve $\sin 2x = \cos x$ for $0 \leq x \leq 2\pi$. **[5]**

18. Solve $\tan 2x = 1$ for $0 \leq x \leq \pi$. **[4]**

19. Prove $\tan 2\theta = \dfrac{2\tan\theta}{1 - \tan^2\theta}$. **[4]**

20. Solve $3\sin x = 2\cos^2 x$ for $0 \leq x \leq 2\pi$. **[5]**

21. Find the exact value of $\sin 75°$. **[4]**

22. Solve $\cos 2x + \sin x = 0$ for $0 \leq x \leq 2\pi$. **[5]**

23. Solve $2\sin x \cos x = \sin x$ for $0 \leq x \leq 2\pi$. **[5]**

24. Express $3\sin x + 4\cos x$ in the form $R\sin(x + \alpha)$. **[5]**

25. Solve $\sin x + \cos x = \sqrt{2}$ for $0 \leq x \leq 2\pi$. **[5]**

End of Worksheet • *Show all working clearly.*